BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 016 · Variables, Input, Output and One-Dimensional Arrays

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. When does ascending bubble sort swap two adjacent items?

Q2. What does CPU cache hold?
A. Only saved personal files
B. Only website passwords
C. Only the next address
D. Often-used data and instructions

Q3. What is a CPU register?

Q4. The next instruction is at address 240. An addition gives 18. What do the accumulator and Program Counter hold? Explain the difference.

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I Do · Read new knowledge

Key vocabulary

variable — Named storage for a value in a program.

assignment — Storing or updating a variable's value.

one-dimensional array — Indexed collection of values stored under one name.

index — Number specifying an array element's position.

New learning

Variables: A variable stores a value that can be updated using assignment. Use meaningful names.

Input and output: Input obtains information from a user or device; output displays or sends information.

Arrays: A one-dimensional array stores related values under one name, accessed using indexes.

Zero-based indexing: Index 0 is the first value; index 1 is the second, so the index differs from counting positions in everyday speech.

Worked example

Trace the array and variable assignments in a short quiz-mark program.

1. Create marks = [4, 7, 9]. The first value has index 0.

2. Set total = marks[0] + marks[1], giving 11.

3. Update marks[0] to 5.

4. Recalculate total = marks[0] + marks[1] to obtain 12.

5. Output total; explain why the value changed.

We Do · Practise with guidance

Guided 1

Write an assignment that stores 12 in a variable called score.

Worked method: score = 12.

Guided 2

For array marks = [6, 3, 8], identify marks[1].

Worked method: 3.

Guided 3

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Update marks[0] to 10 and calculate marks[0] + marks[2].

Worked method: 10 + 8 = 18.

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You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write one print instruction for the updated value.

```
fileMB = 5
fileMB = fileMB + 2
```

write

Question 1

State one difference between a variable and an array. [2 marks]

Question 2

Given values = [7, 2, 5], state the values at indices 0 and 2. [2 marks]

Question 3

Write OCR ERL assignments to set total to 0, add values[0] and values[1], then display total. [4 marks]

Exit Ticket · Answer before checking

The element at index 2 is normally the third element of an OCR ERL array.

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GREEN PEN Then: tick and rewrite mistakes.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. When the left item is greater than the right item.

Q2. D — Often-used data and instructions

Q3. A small, fast storage location inside the CPU. It holds data or addresses in use.

Q4. The accumulator holds the result, 18. The PC holds the next instruction's address, 240.

I Do / We Do pages · Guided practice

Q1. score = 12.

Q2. 3.

Q3. 10 + 8 = 18.

You Do pages · Code First

```
fileMB = 5
fileMB = fileMB + 2
print(fileMB)
```

Expected result: 7

You Do pages · Independent questions

Q1. Variable can store a value [1]; array contains an indexed collection of values [1].

Q2. Index 0 is 7 [1]; index 2 is 5 [1].

Q3. total = 0 [1]; total = total + values[0] [1]; total = total + values[1] [1]; print(total) [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: